

Customer Churn Prediction & Intelligent Retention System

Executive & Technical Report — Machine Learning Pipeline for Proactive Customer Retention

Executive Summary

Customer churn — the loss of subscribers to competitors — is one of the most direct threats to recurring revenue in the telecommunications industry. This project delivers a complete, production-ready machine learning pipeline that identifies customers at high risk of churning before they leave, giving the business a window to intervene with targeted retention offers. Using the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes), we engineered predictive features, benchmarked six modeling approaches, and selected **XGBoost (Tuned)** as the champion model, achieving a **ROC-AUC of 0.846** and **Recall of 81.5%** on held-out test data — meaning the model correctly flags roughly 4 out of 5 customers who are about to churn.

Business Problem Statement

The dataset shows an overall churn rate of approximately 26.5%, representing a substantial and continuous drain on recurring revenue. Acquiring a new customer is well documented to cost significantly more than retaining an existing one, making early identification of at-risk customers a high-leverage lever for improving profitability. Without a predictive system, retention efforts are either applied blanket-wide (expensive and inefficient) or reactively, after a customer has already decided to leave (too late to act). This project closes that gap by scoring every customer with a quantified churn probability, enabling the business to prioritize retention spend on the customers who need it most.

Key Business Findings

- Month-to-month contracts churn at far higher rates than one- or two-year contracts, identifying contract flexibility as the single largest churn lever.
- Churn risk is concentrated in the first 12 months of the customer lifecycle — the critical window for onboarding and early engagement.
- Customers paying via Electronic Check churn more than those on automatic payment methods, a proxy for lower commitment/engagement.
- Customers without add-on services (Online Security, Tech Support, etc.) churn more, suggesting these services increase switching costs and perceived value.
- High Monthly Charges combined with short tenure and month-to-month contracts defines the highest-risk customer segment in the portfolio.

Dataset & ML Methodology

Dataset: IBM Telco Customer Churn — 7,043 customers, 21 raw attributes spanning demographics, account information, and subscribed services. **Data Hygiene:** The *TotalCharges* field was coerced from string to numeric, with 11 blank-string records (all new customers with zero tenure) imputed to 0.

Feature Engineering: Four new features were constructed to capture engagement and risk signals beyond the raw schema: *Total_Services_Used* (count of subscribed add-ons), *Estimated_LTV* (MonthlyCharges × tenure), *Tenure_Group* (lifecycle-stage bucket), and *Payment_Risk_Score* (flag for manual/non-automatic payment

methods).

Preprocessing: Binary target encoding (Churn), one-hot encoding for nominal categorical variables, standard scaling for continuous numeric features, and an 80/20 stratified train-test split (random_state=42) to preserve the class balance in both partitions.

Modeling: A Logistic Regression baseline was benchmarked against Decision Tree, Random Forest, XGBoost, and SVM classifiers. Random Forest and XGBoost were further optimized via RandomizedSearchCV with 5-fold Stratified cross-validation, tuned for ROC-AUC.

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost (Tuned)	0.7410	0.5075	0.8155	0.6256	0.8456
Gradient Boosting	0.8013	0.6577	0.5241	0.5833	0.8447
Random Forest (Tuned)	0.7516	0.5212	0.7888	0.6277	0.8436
Logistic Regression (Baseline)	0.7999	0.6554	0.5187	0.5791	0.8424
Decision Tree (GridSearchCV)	0.7942	0.6312	0.5401	0.5821	0.8267
Random Forest (Default)	0.7885	0.6258	0.5053	0.5592	0.8234
Decision Tree	0.8006	0.6667	0.4973	0.5697	0.8213
XGBoost (Default)	0.7708	0.5780	0.5053	0.5392	0.8120
SVM (RBF)	0.7984	0.6654	0.4840	0.5604	0.7979

The champion model (XGBoost (Tuned)) is highlighted above, selected on the combination of ROC-AUC and Recall — the two metrics most aligned with the cost asymmetry of missed churners versus wasted retention offers (see Cost-Benefit discussion below).

Top 10 Feature Importance Findings

Rank	Feature	Importance
1	Contract_Two year	0.2159
2	Contract_One year	0.1880
3	InternetService_Fiber optic	0.0866
4	OnlineSecurity_No internet service	0.0710
5	InternetService_No	0.0510
6	tenure	0.0477
7	PaymentMethod_Electronic check	0.0456
8	StreamingMovies_Yes	0.0322
9	PaperlessBilling	0.0251
10	OnlineSecurity_Yes	0.0212

Contract type dominates feature importance, followed by internet service type, tenure, and payment method — confirming that commitment structure and customer lifecycle stage are the primary churn drivers, ahead of demographic factors.

Cost-Benefit Rationale: Why Recall & ROC-AUC

A **False Positive** (flagging a loyal customer as a churn risk) costs the business a modest, bounded amount — an unnecessary discount or outreach call. A **False Negative** (missing an actual churner) costs the business the customer's full remaining lifetime value plus the higher expense of acquiring a replacement. Because missed churners are far more expensive than unnecessary offers, the pipeline prioritizes **Recall** — catching as many true churners as possible — while using **ROC-AUC** to select the model with the best overall ranking quality across all decision thresholds, rather than optimizing for raw Accuracy, which can be misleading on this moderately imbalanced (~26.5% churn) dataset.

5 Actionable Retention Strategies

1. Personalized Retention Offers

Route customers with a churn probability above the high-risk threshold (e.g., 70%+) to a personalized save offer — a discount, plan credit, or service upgrade tailored to their usage pattern and current contract type — rather than a generic, one-size-fits-all promotion.

2. Loyalty & Contract Migration Rewards

Since month-to-month customers churn at the highest rate, offer incentives (bill credits, free add-on trials) for migrating to one- or two-year contracts, directly targeting the single strongest churn driver identified by the model.

3. Early Churn Alert System

Operationalize the model to score every customer monthly and trigger automated alerts to the retention team when a customer crosses into the medium- or high-risk band, enabling proactive outreach before the customer initiates a cancellation.

4. Engagement & Onboarding Campaigns

Since churn risk peaks in the first 12 months, invest in a structured onboarding journey (welcome calls, usage tutorials, milestone check-ins) for new customers to build habitual engagement during the highest-risk window.

5. Service Upgrade & Bundling Incentives

Customers without add-on services (Online Security, Tech Support, Device Protection) churn more; proactively offering bundled add-ons at a discount increases both switching costs and perceived value, particularly for high-Monthly-Charge fiber customers.

Strategic Conclusion

The XGBoost (Tuned) model provides the business with a reliable, quantified early-warning system for customer churn, correctly identifying approximately 82% of at-risk customers before they leave. Deploying this model as an ongoing scoring service — paired with the five retention strategies above — creates a closed-loop system: predict risk, intervene proactively, and measure the resulting change in churn rate over time. Suggested next steps include A/B testing retention offers against model-identified risk segments, integrating the scoring pipeline into the CRM for real-time alerts, and periodically retraining the model as customer behavior and market conditions evolve.

Report generated automatically from model artifacts (trained 2026-08-31 22:44:47). All metrics reflect performance on a held-out 20% test set never seen during training or hyperparameter tuning.